

ALI BASHAR ALABID

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PROFESSIONAL SUMMARY

Computer Science student, AI & Digital Heritage Researcher, and Head of Operations with proven expertise in full-stack software development, computer vision applications, and large-scale digital preservation. Experienced in leading multidisciplinary technical teams (from 4 to 21 members) to deliver international research, interactive virtual experiences, and automated systems. Proficient in leveraging PHP, MySQL, and AI-assisted workflows to build high-impact, production-ready solutions.

EDUCATION

Bachelor's Degree in Computer Science | Al-Noor University — Mosul, Iraq

Expected Graduation: July 2027

- Focus areas: Software Engineering, Computer Vision Applications, and Digital Heritage Technologies.

PROFESSIONAL EXPERIENCE

Head of Operations | Al-Noor Center for Digital Heritage — Mosul, Iraq

Official Website : <https://center.alnoor.edu.iq/en>

- Coordinate operations and deliverables across software development, AI, 3D digitization, and digital preservation teams.
- Manage cross-functional collaboration between technical and non-technical teams to ensure strict alignment with project timelines.
- Streamline workflows for digital documentation, interactive virtual experiences, and high-fidelity heritage preservation outputs.

PROJECT EXPERIENCE

Al-Noor Library Management System | *Full-Stack Developer*

- *Technologies:* PHP, MySQL, NFC, QR Codes, PHPMailer.
- Designed and deployed a full-stack university library system, integrating automated NFC identification and automated borrowing workflows.
- Successfully tested the system with a pilot group of 150 active users before final deployment.
- Built and optimized the database infrastructure to scale seamlessly from an initial 200-book catalog to over 5,000 fully indexed records managed by the head librarian.
- Implemented role-based access control and local network multi-device synchronization.

SmartSignal-AI (Adaptive Traffic Signal Control) | *Team Leader & AI Integration Lead*

- *Technologies:* Python, YOLO, Computer Vision, Systems Optimization.
- Led a technical team of 4 members to develop an AI-powered adaptive traffic management system.

- Trained, validated, and deployed a **YOLO** computer vision model for real-time lane occupancy detection and traffic density estimation.
- Engineered communication workflows between real-time camera monitoring systems and adaptive signal decision logic.
- Integrated emergency vehicle prioritization rules to dynamically optimize traffic signal timings.

Immersive & AI-Generated Virtual Tours | *Lead Developer*

- *Technologies:* 360° Photogrammetry, AI Image Generation, Panorama-Based Visualization.
- Developed the official Al-Noor University Virtual Tour generating an average of 50 active monthly visits. Live Demo: virtualtour.alnoor.edu.iq
- Designed and built interactive virtual tours for the Mosul Museum and Baitna Foundation to enable remote exploration and digital heritage preservation.
Live Demo: <https://bytna.vercel.app>
- Spearheaded an experimental immersive environment leveraging AI-generated imagery to showcase advanced applications of generative AI in digital spaces.
Live Demo: <https://babl360.vercel.app>

Discover Mosul Application | *Head of Content Collection Team*

Live Web App : <https://yalla-mosul.web.app>

- Led a multidisciplinary team of 21 members tasked with collecting, structuring, and verifying regional tourism and heritage data.
- Established data validation and QA frameworks to build a comprehensive digital information repository for historical landmarks.

Epic of Gilgamesh Film Trailer | *Visual Content Creator & Storyboard Designer*

- Collaborated within a creative production team to design and produce a cinematic concept trailer for the "Epic of Gilgamesh" film.
- Spearheaded visual asset generation, selecting and compositioning high-quality thematic imagery to accurately represent historical narratives.
- Engineered the visual pacing and continuity, establishing seamless transitions and logical connections between shots to maintain storytelling flow.
- **Watch Trailer:** <https://youtu.be/04HkIFVki94?si=GSoaeksbdCj-RApy>

RESEARCH & PUBLICATIONS

Photogrammetry-Based Digital Twin for Mar Mattai Monastery | *Scopus-Indexed Publication*

- Published co-authored research in a Scopus-indexed venue focusing on 3D digitization and digital preservation. View Publication: https://doi.org/10.18420/inf2025_60
- Applied advanced photogrammetry and 3D reconstruction techniques to build a highly accurate digital twin of the historic monastery.
Project Showcase : <https://pih.education/projects>

LEADERSHIP & INTERNATIONAL ACTIVITIES

- **Informatik Festival 2025 (Germany):** Represented regional digital heritage and AI technology initiatives at a premier international computer science event.
- **AI Heritage Exhibition (Team Lead):** Guided a team of 7 members to orchestrate an exhibition featuring the restoration and animation of 24 historical artworks using AI tools.
- **Archaeological Tourism Conference (Organizer & Mentor):** Mentored 12 students in content creation and presentation readiness for historical posters focused on local archaeology and heritage.

AWARDS & ACHIEVEMENTS

- **Best Student Poster Award** – Digitalization of Cultural Heritage Conference, Cyprus (2025).
- **Published Author** – Scopus-indexed research publication in digital cultural heritage.
- **National Rank (8th Place)** – Iraqi Innovation & Creativity Challenge.
- **Academic Training:** Completed international academic training in Digitalization of Cultural Heritage via Technische Hochschule Brandenburg.

TECHNICAL SKILLS

- **Programming:** PHP, Python, JavaScript, C#, Java, SQL, HTML, CSS.
- **Databases:** MySQL.
- **Artificial Intelligence:** YOLO Model Training, Computer Vision, AI Workflow Automation, Generative AI Integration.
- **Digital Heritage:** 360° Photogrammetry, 3D Reconstruction, Digital Twin Development, Virtual Tours.
- **Tools & Platforms:** Git, GitHub, WAMP Server, Microsoft Office, Google Workspace.

LANGUAGES

- **Arabic:** Native.
- **English:** Professional Working Proficiency.